الجامعة التكنولوجية

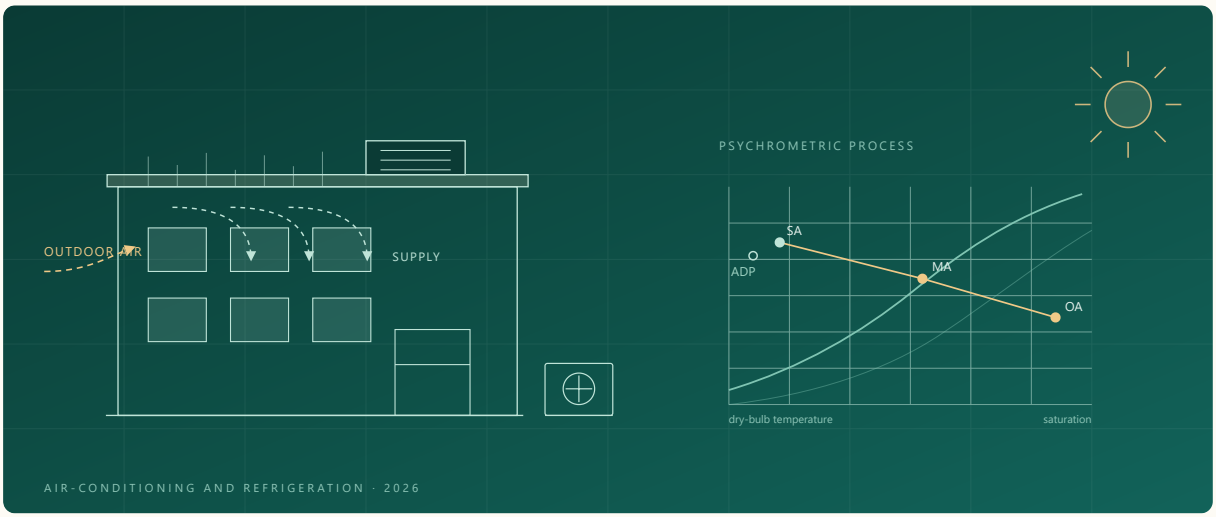
كلية الهندسة الكهروميكانيكية

قسم النظم والاتمتة

PROJECT REPORT · 2026

HVAC Design Studio

An integrated platform for cooling-load, air-handling, refrigeration and comfort-control education



COURSE INFORMATION

اسم المادة	أنظمة التكييف والتبريد
المرحلة	الرابعة
إشراف	الدكتورعدنان

STUDENTS

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من ستة أسئلة إلى نظام مكيّف كامل — ومن الحمل إلى المحاكاة.

PREPARED TO ACCOMPANY THE LIVE DEMONSTRATION

Abstract

INSTITUTION
University of Technology

DEPARTMENT
Systems and Automation Engineering

COURSE
Air-Conditioning and Refrigeration Systems

HVAC Design Studio is a web-based engineering workspace that connects four calculations usually taught apart: peak cooling load, moist-air processes in an air-handling unit, the real-fluid vapour-compression cycle, and closed-loop comfort control. One 96 m² classroom case is carried through the whole chain: 12.83 kW sensible and 1.19 kW latent room gains, plus 5.13 kW of outdoor-air load, become a **19.16 kW** design load (5.45 TR); the AHU produces 1.140 kg/s of 13 °C supply air and a **19.17 kW** coil duty at 91.6% SHR; an R134a cycle at 2/40 °C converts that duty into a **COP of 3.976**; and a PID loop holds **24.50 °C** with 100% final-half comfort.

A six-answer studio at the front runs the same load engine without the full input set: the default office needs **18.83 kW**, receives a **25 kW VRF multi-split** (2 × 12.5 kW), and is simulated through a 24-hour design day with pre-cooling, night setback, capacity limits and an ambient/part-load COP correction. The interface is bilingual (Arabic RTL / English), offers light and dark themes, and is supported by 92 automated checks and a browser screenshot matrix.

The report documents the method, the governing equations, the interface, the numerical results and the limits of the model, so that every number on the screen can be traced back to an assumption or a measurement.

KEYWORDS cooling load · psychrometrics · air-handling unit · vapour compression · R-134a · PID control · VRF · design-day simulation · engineering education

QUICK STUDIO LOAD
18.83 kW
Six-answer office case

DETAILED DESIGN LOAD
19.16 kW
5.45 TR · RSHF 91.5%

COIL DUTY
19.17 kW
SHR 91.6% · BPF 0.95

CYCLE COP
3.976
R134a · Carnot 6.399

FINAL ROOM CONDITION
24.50 °C
39.9% RH · 100% comfort

DESIGN-DAY ELECTRICITY
72.8 kWh
190.9 kWh cooling · COP 2.62

The report is deliberately short: four physical models, one six-answer studio, one design day, and a plain statement of the model's limits. Each section carries the result that leaves it and enters the next.

CAPABILITIES INCLUDED

- Peak load with component audit and RSHF
- Moist-air states, mixing, ADP and BPF
- R134a cycle with CoolProp states
- PID and threshold comfort control
- Six-answer quick studio and sizing rule
- 24-hour design day at 10-minute steps
- Arabic / English and light / dark themes
- 92 automated checks and browser QA

01

INTRODUCTION

Four chapters, one connected case.

An air-conditioning design is a chain of dependent decisions. This project builds a workspace that makes the chain visible, calculable and presentable.

1.1 Background and motivation

Undergraduate HVAC courses are organised by topic. Cooling-load estimating, moist-air properties, air-handling equipment, the vapour-compression cycle and automatic control each occupy their own chapter, with their own symbols and their own worked examples. The arrangement is efficient for teaching, but it hides the property that matters most in practice: **the output of one calculation is the input of the next**.

A student can compute a room load, a coil duty, a COP and a PID response in four separate exercises and still be unable to answer a simple question: which number, on this screen, is the one that sizes the equipment? The project closes that gap with a single case. A 96 m^2 classroom is defined once; its room gains become a design load, the load becomes a supply-air state, the state becomes a coil duty, the duty becomes a refrigeration cycle, and the cycle efficiency becomes the electricity line in a comfort simulation.

1.2 Objectives

01 Implement a transparent peak-load engine that reports every component and the room sensible heat factor.

02 Turn the room load into a supply-air state and a coil duty, including mixing, apparatus dew point and bypass factor.

03 Solve the real-fluid vapour-compression cycle and compare it with the Carnot ceiling at the same saturation temperatures.

04 Close a comfort loop around a reduced-order zone model with PID and on/off control, reporting energy and actuator effort.

05 Unify the four layers into one bilingual workspace, with a six-answer front end and a 24-hour design day.

1.3 Scope and approach

The platform is a **teaching and preliminary-design tool**. Its load calculation is a steady peak estimate, and its design day is a reduced-order dynamic model. Neither is a construction-issue package, an annual energy simulation or a manufacturer's equipment selection. The thermodynamic and load calculations live in a small Python package with no interface code; the Streamlit layer only collects inputs, calls the engines and presents results, so the physics is testable independently of the browser.

Duct pressure loss, fan-system curves, annual weather, multi-zone coupling, construction drawings, cost and equipment selection are outside the scope; Section 7.2 lists the limits and their effect on the numbers.

> *"The value of the tool is not a single number. It is the visible hand-off between numbers — and the ability to defend each hand-off in front of a class."*

02

THE DESIGN CASE AND THE PLATFORM

One classroom, one set of numbers.

A single Iraqi summer case anchors every module. The assumptions are ordinary engineering values; what is unusual is that they are used consistently everywhere.

2.1 The classroom case

The case is a 12.0 × 8.0 × 3.2 m classroom: 96 m^2 of floor, 307.2 m^3 of volume, with 80% of the perimeter exposed. The envelope is a 200 mm brick wall with 50 mm insulation, a concrete roof with 75 mm insulation, and 18 m^2 of double low-e glazing. The main glazing faces west, which places the solar peak in the late afternoon, when the outdoor air is already hottest. The design day is 45 °C dry-bulb and 24 °C wet-bulb; the room is held at 24 °C and 50% RH. Twenty occupants work at office activity, with 10 L/s of fresh air per person and 0.50 air changes per hour of infiltration.

Parameter	Default value	Basis / note
Geometry	12.0 × 8.0 × 3.2 m	96 m^2 · 307 m^3 · 80% exposed
Wall	Brick + 50 mm insulation · U 0.55	α 0.65 · 600 W/m^2 · film 20
Roof	Concrete + 75 mm insulation · U 0.35	α 0.75 · 950 W/m^2
Glazing	Double low-e · 18.0 m^2 · U 1.60	SHGC 0.40 · shading 0.85
Outdoor design	45.0 °C DB / 24.0 °C WB	Hot-dry · 101.325 kPa
Indoor design	24.0 °C / 50% RH	Setpoint used by all four modules
Occupancy	20 people · office work	75 W + 55 W per person
Internal gains	8 W/m^2 lighting · 1.5 kW equipment	Connected equipment load
Ventilation	10 L/s per person	720 m^3/h of outdoor air
Infiltration	0.50 ACH	Outdoor air through the envelope

Table 1 — Default design case. Every value is editable in Module 1. The quick studio derives an equivalent set of inputs from six simpler answers, so the two paths can be compared without changing the physics.

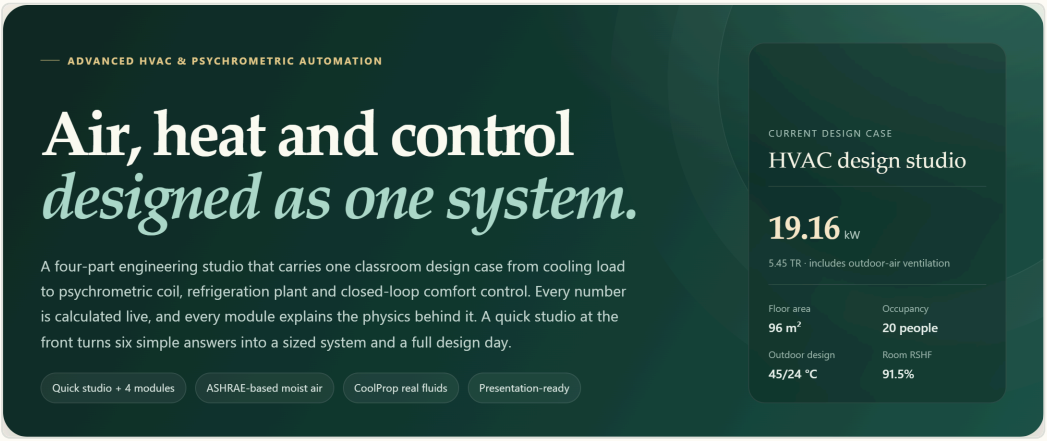


Figure 1 — The overview workspace. The landing page states the connected case, shows the live load and the four module results, and provides the route to the quick studio. The sidebar carries the current case, the language selector and the theme toggle.

The case is intentionally *mostly sensible*: the room sensible heat factor is 91.5%. That makes it a good teaching vehicle for the difference between room RSHF and coil SHR, and it reflects a real classroom in Baghdad, where envelope and solar gains dominate.

03

CHAPTER THREE

The engineering behind the screen.

Four physical models — load, air, refrigerant and control — each solved in SI units and traceable to a standard textbook equation.

3.1 Cooling load

The load model follows the ASHRAE cooling-load procedure at design conditions rather than a rule-of-thumb area factor. Heat enters through the envelope, is generated inside, and is carried in with outdoor ventilation air. Opaque surfaces use the sol-air temperature, which folds absorbed solar radiation into an equivalent outdoor air temperature; glazing uses a conduction term plus a solar term, because a window admits heat by both routes and the second usually dominates in Baghdad.

$$Q_{op} = \Sigma\, U\, A\, (t_{sol} - t_i) \quad t_{sol} = t_o + \alpha\, I / h_o \tag{1}$$

$$Q_{gl} = \Sigma\, [U\, A\, (t_o - t_i) + \mathrm{SHGC} \cdot A\, I] \tag{2}$$

Internal gains follow the same audit: people, lighting, equipment, infiltration and ventilation are separate rows in Table 2, and SHR and RSHF are formed from their sums.

Component	Sensible kW	Latent kW	Total kW	Share
External wall, 84.4 m² at U = 0.55	1.880	—	1.880	9.8 %
Roof, 96 m² at U = 0.35	1.903	—	1.903	9.9 %
Window conduction, 18 m² at U = 1.60	0.605	—	0.605	3.2 %
Window solar gain, SHGC 0.40, shading 0.85	3.672	—	3.672	19.2 %
Occupants, 20 people	1.500	1.100	2.600	13.6 %
Lighting, 8 W/m²	0.768	—	0.768	4.0 %
Equipment	1.500	—	1.500	7.8 %
Infiltration, 0.5 ACH	1.004	0.091	1.095	5.7 %
Room load	**12.832**	**1.191**	**14.022**	**73.2 %**
Outdoor-air ventilation, 10 L/s/person	4.707	0.426	5.133	26.8 %
Design cooling load	**17.539**	**1.617**	**19.155**	**100 %**

Table 2 — Cooling-load components at design conditions. 19.155 kW is 5.447 tons of refrigeration. Sensible heat ratio 91.56 %; room sensible heat factor 91.51 %. Shares are of the design total, so the ventilation row is not additive with the room total.

Two consequences follow. Envelope and solar gain together contribute 8.06 kW, 42 % of the design total, so glazing and roof insulation dominate any efficiency measure. The 1.62 kW latent share is small but decisive: it fixes the supply-air humidity and the coil's dew point.

DESIGN LOAD
19.16 kW
5.45 tons of refrigeration

ROOM SENSIBLE
91.5 %
RSHF; coil SHR 91.6 %

ENVELOPE + SOLAR
8.06 kW
42 % of the design total

VENTILATION
5.13 kW
26.8 % of the design total

3.2 Psychrometrics and the air-handling unit

A cooling load is a heat rate; an air-handling unit converts it into a volumetric flow. The bridge is moist air. The model evaluates every airstream from dry-bulb temperature and relative humidity using the ASHRAE formulations, with saturation pressure from the Hyland–Wexler correlation at ambient pressure. Outdoor air at 19.16 % of the total mass flow mixes with return air before the coil; the mixture is cooled and dehumidified towards the apparatus dew point, and the bypass factor sets how far the supply air lands along that line.

$$W = 0.622\, p_w / (p - p_w) \quad h = 1.006\, t + W\,(2501 + 1.86\, t) \tag{4}$$

$$\dot{m}_a = \dot{V} / v \quad Q_{coil} = \dot{m}_a\,(h_{ma} - h_{sa}) \tag{5}$$

$$\mathrm{BPF} = (h_{sa} - h_{adp}) / (h_{ma} - h_{adp}) \tag{6}$$

State point	t (°C)	t_{wb} (°C)	RH (%)	W (g/kg)	h (kJ/kg)	v (m³/kg)
Outdoor air (OA)	45.00	24.00	16.8	10.08	71.3	0.916
Return air (RA)	24.00	17.07	50.0	9.30	47.8	0.854
Mixed air (MA)	28.03	18.55	40.0	9.45	52.3	0.866
Supply air (SA)	13.00	12.56	95.2	8.88	35.5	0.822
Apparatus dew point	12.21	12.21	100	8.85	34.6	0.820

Table 3 — Moist-air state points. Computed at sea-level pressure. Supply air at 13 °C and 95 % RH is deliberately close to saturation, which is what a wet cooling coil produces. The mixed row is the mass-weighted mix: 0.1916 × 10.08 + 0.8084 × 9.30 = 9.45 g/kg.

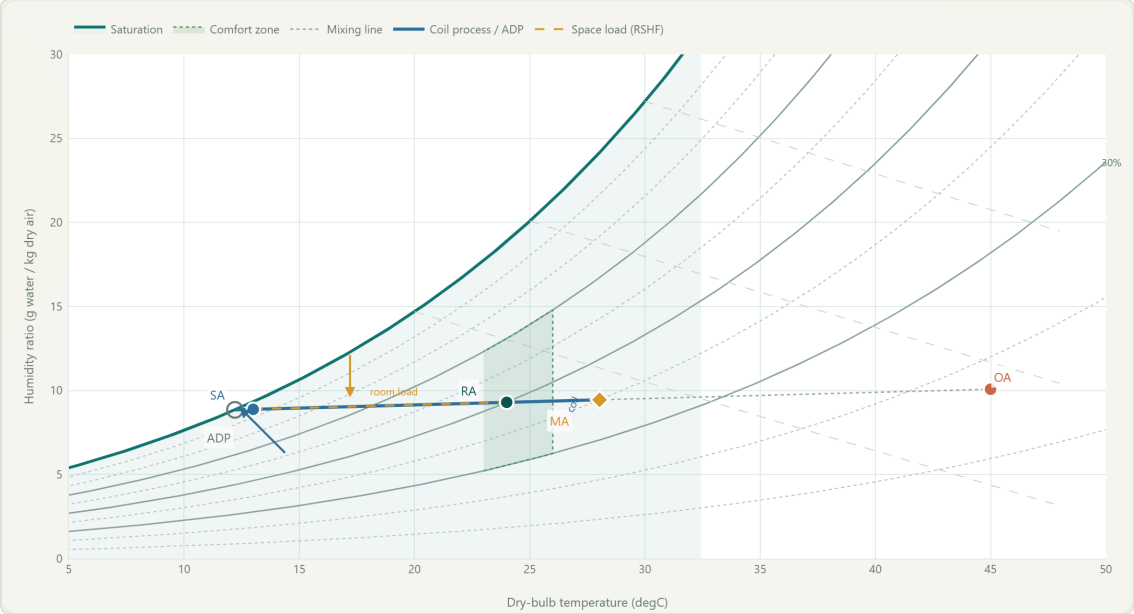


Figure 2 — Psychrometric process of the air-handling unit. A live capture from Module 2. The process line runs from the mixed-air point through the coil towards the apparatus dew point; the supply-air marker is held short of the saturation curve by the 0.95 bypass factor.

COIL DUTY
19.17 kW
17.55 sensible + 1.62 latent

SENSIBLE RATIO
0.916
high-sensible, dry-climate load

BYPASS FACTOR
0.950
ADP 12.21 °C at the coil face

AIR MASS FLOW
1.140 kg/s
0.937 m³/s ≈ 3 370 m³/h

CLOSING THE ENERGY BALANCE

Table 3 can be checked without the code. Enthalpy: 1.140 × (52.3 − 35.5) = 19.15 kW against a 19.17 kW coil duty. Moisture: 1.140 × 2 500 × (9.45 − 8.88) / 1 000 = 1.62 kW. The 0.02 kW gap is rounding, not a second model.

3.3 Vapour-compression cycle

Module 3 takes the coil duty from Module 2 and solves an ideal R134a cycle with a real-fluid property library. Four state points characterise the loop: compressor suction, compressor discharge, condenser outlet and evaporator inlet. Expansion is isenthalpic; compression is not isentropic, and the isentropic state is computed alongside the real one so that the isentropic efficiency can be shown.

$$\mathrm{COP} = (h_1 - h_4) / (h_2 - h_1) \tag{7}$$

$$\mathrm{COP}_{\mathrm{Carnot}} = T_{\mathrm{L}} / (T_{\mathrm{H}} - T_{\mathrm{L}}) \quad \eta_{\mathrm{II}} = \mathrm{COP} / \mathrm{COP}_{\mathrm{Carnot}} \tag{8}$$

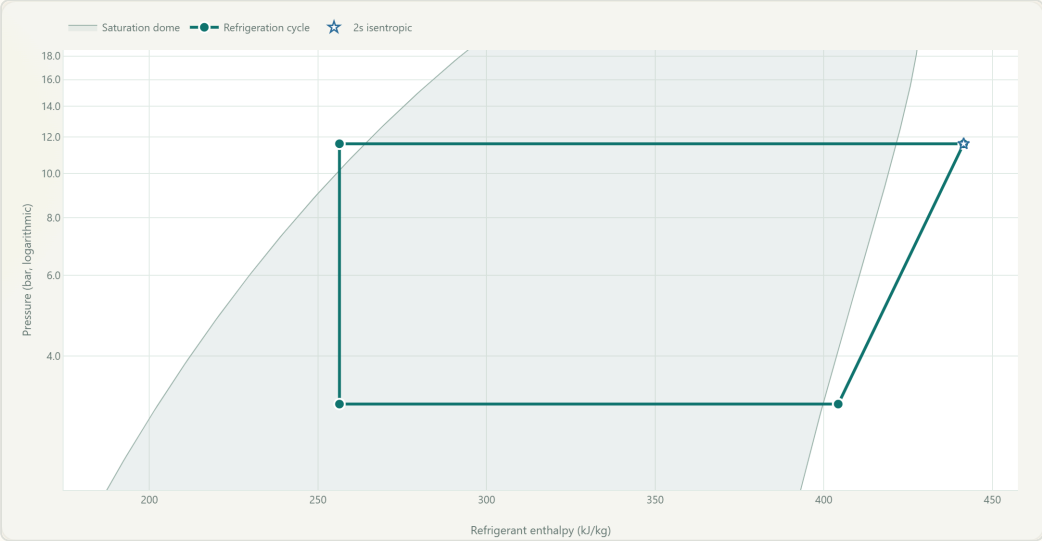


Figure 3 — Pressure–enthalpy diagram of the R134a cycle. Saturation dome in grey, cycle in teal. The compressor raises pressure from 3.15 to 11.60 bar; the condenser rejects 23.99 kW at 40 °C; the expansion valve drops the refrigerant back to 2 °C at constant enthalpy.

Point	Description	p (bar)	t (°C)	h (kJ/kg)	s (kJ/kg·K)
1	Compressor suction, saturated vapour	3.146	7.00	404.28	1.7422
2	Compressor discharge, superheated	11.599	62.65	441.47	1.7703
2s	Isentropic discharge, reference	11.599	54.24	432.17	1.7422
3	Condenser outlet, saturated liquid	11.599	40.00	256.38	1.1900
4	Evaporator inlet, after expansion	3.146	2.00	256.38	1.2049
PERFORMANCE AT THE DESIGN COIL DUTY OF 19.17 KW					
Refrigerant mass flow				0.1296 kg/s · specific refrigeration 147.90 kJ/kg	
Compressor work				4.820 kW · condenser 23.987 kW · pressure ratio 3.687	
Coefficient of performance				**COP 3.976 · Carnot 6.399 · second-law efficiency 62.1 %**	

Table 4 — Cycle states and performance. Properties evaluated with CoolProp for R134a. An isentropic efficiency of 0.75 explains the gap between points 2 and 2s; the 62 % second-law efficiency is the share of reversible work that the real compressor achieves.

At the design duty the refrigerant mass flow is 0.1296 kg/s, compressor work is 4.820 kW and the condenser rejects 23.987 kW. A COP of 3.976 against a Carnot ceiling of 6.399 is a second-law efficiency of 62.1 % — the honest measure of how far the real machine sits from the reversible limit.

> **WHY 2 °C AND 40 °C**
>
> The saturation temperatures are not arbitrary. The 2 °C evaporator is cold enough to dehumidify the 13 °C supply air; the 40 °C condenser is hot enough to reject heat to a 46 °C day with a finite condenser approach. Raising either temperature lowers the pressure ratio, but it also moves the cycle away from the coil duty it must serve.

3.4 Comfort and automatic control

The last module closes the loop. A room model receives the cooling delivered by the plant, loses heat through the same envelope as Module 1, gains heat from the same internal sources, and reports a zone temperature every minute. The controller forms the error as room minus setpoint — positive when cooling is needed — and commands a valve position between zero and one. Comfort is assessed on ET* and relative humidity: 23–26 °C and 30–70 % RH.

$$e(t) = T_z(t) - T_{sp} \quad u(t) = K_p\, e + K_i \int e\, dt + K_d\, de/dt \tag{9}$$

$$\text{comfort: } 23\ °C \le ET^* \le 26\ °C \ \text{ and } \ 30\ \% \le RH \le 70\ \% \tag{10}$$

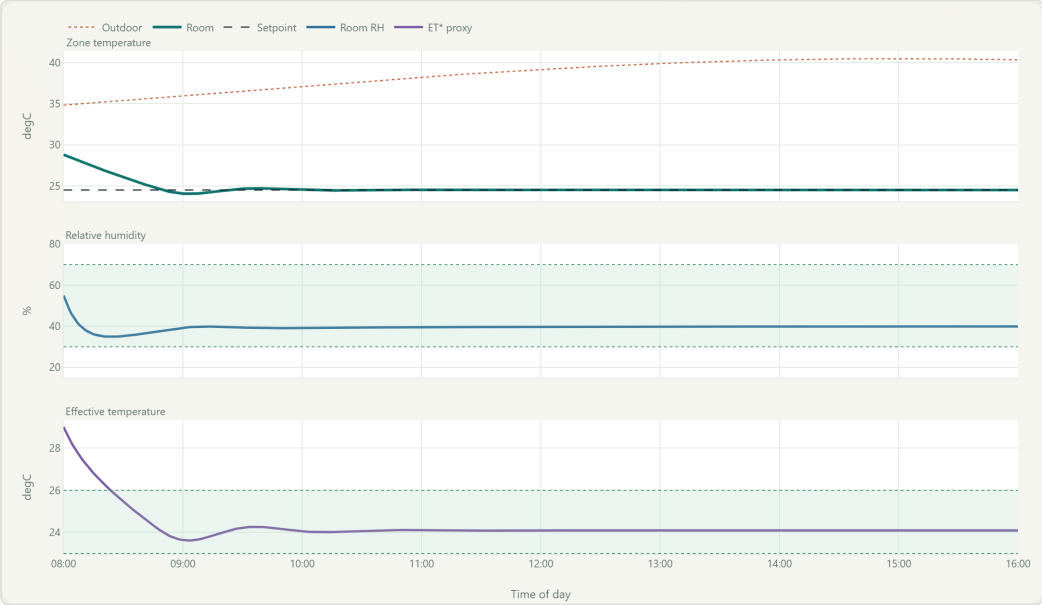


Figure 4 — Comfort timeline. Zone temperature, setpoint, outdoor temperature, humidity and ET* from 08:00 to 16:00 with the PID controller. The room starts near 29 °C, is pulled down to 24 °C within the first hour and then held flat while the outdoor temperature climbs above 40 °C.

Indicator	PID	Two-position	Reading
Final zone temperature	24.498 °C	24.352 °C	Both within 0.5 K of setpoint
Final relative humidity	39.90 %	39.89 %	Identical dehumidification
Comfort, final half	100 %	100 %	Neither leaves the band
Cooling delivered	73.20 kWh	73.25 kWh	0.07 % difference
Electricity used	18.41 kWh	18.42 kWh	0.06 % difference
Switch changes	**0**	**55**	**Wear and noise, not energy**

Table 5 — PID versus two-position control over the eight-hour run. The plant is the same in both runs. The only difference is the control law, so any change in energy can be attributed to control and not to equipment.

The honest conclusion is that the two-position controller is not wasteful in energy — it is wasteful in actuator life. On a small packaged unit with an on/off compressor that distinction matters less; on a modulating valve it decides maintenance cost.

> **WHY A DEADBAND**
>
> A two-position controller switches the valve fully on and fully off; in the comparison run it produced 55 switch changes in a single day. A PID controller modulates continuously — fully open for the morning pull-down, then near three-quarters open — with zero switch changes at almost the same energy. All gains are editable in Module 4.

04

CHAPTER FOUR

Six answers. A sized system.

A visitor who knows only the space, not the physics, can obtain a defensible load, a recommended system and a full design day in under a minute.

4.1 The six answers

The studio asks exactly six questions. Everything else — internal gains, ventilation rate, infiltration, wall and roof construction, occupancy — is filled from a named engineering preset, and every preset is visible and editable in the advanced panel.

Question	Default answer and what it sets
Type of space	Office — occupancy density, lighting, plug load, ventilation and schedule
Floor area	96 m² — with the office preset this implies 11 occupants
City	Baghdad, 46 °C DB / 24 °C WB; other Iraqi cities carry their own design day
Glazing ratio	25 % of the wall area, the default for a normally glazed office
Main orientation	West — the worst façade in Baghdad, where the solar peak meets the air peak
Envelope quality	Standard — insulated wall, double glazing, controlled infiltration

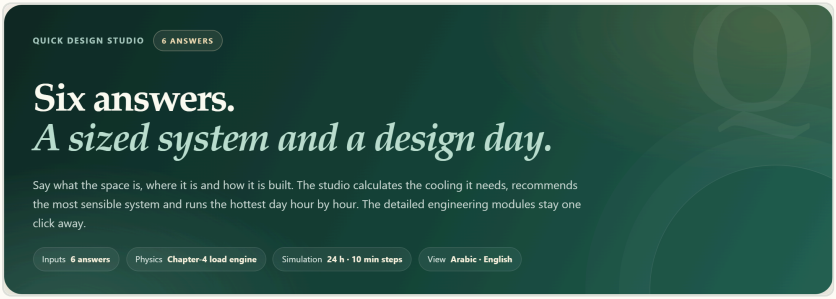


Figure 5 — The quick design studio. The landing panel states the contract in one sentence: six answers in, a sized system and a design day out.

System selection is a rule, not a preference. If the designer has chosen a family, that choice is respected. If the choice is *Auto*, the target load and plan type decide the family: small loads go to high-wall splits, medium enclosed loads to VRF, open-plan loads to ducted packaged units, and large loads to chilled water.

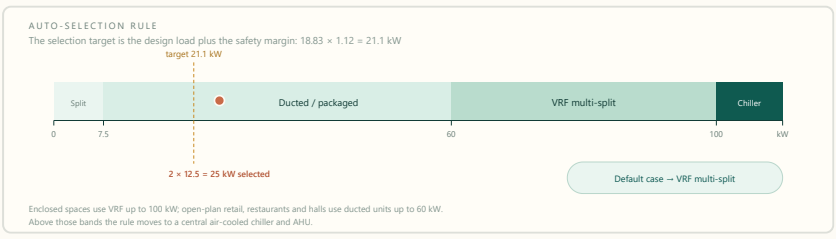


Figure 6 — Automatic system selection. The bands apply to the design target, not the raw load: 7.5 kW divides split from packaged equipment, 60 kW divides packaged from VRF, and above 100 kW the plant becomes chilled water.

MARGIN BEFORE CAPACITY, NOT AFTER

The 12 % margin is applied to the load before a size is chosen, so equipment is compared with 21.09 kW rather than 18.83 kW — the unit is not chosen at the calculated load and then found short once duct gain, fan heat and a hot condenser day take their share.

4.2 From answers to a system

The six answers are expanded into a full input set and passed to *calculate_thermal_load*, the same engine used by Module 1. With 96 m², standard construction, 25 % west-facing glazing and the Baghdad design day the result is 18.83 kW, or 5.36 tons, with 11 occupants. The smallest nominal pair covering the 21.09 kW target is 2 × 12.5 kW.

Result	Value	Basis
Cooling load	18.83 kW · 5.36 TR	Same load engine as Module 1
Selection target	21.09 kW	Load plus the 12 % margin
Recommended system	VRF multi-split	Enclosed, medium load, long day
Outdoor units	2 × 12.5 kW	Smallest nominal pair covering the target
Indoor units	4	One per zone, sized with the outdoor pair
Installed capacity	25.0 kW · 7.1 TR	Nominal, at rating conditions
Supply airflow	6 150 m³/h	From the sensible load and a 9 K ΔT
Fresh air · rated COP · margin	**396 m³/h · 3.65 · 18.5 %**	**10 L/s per person, 11 occupants**

Table 6 — Quick studio result. The exact output of the default six answers. Airflow and fresh air are shown because they constrain the duct and the façade louvre before any equipment is ordered.

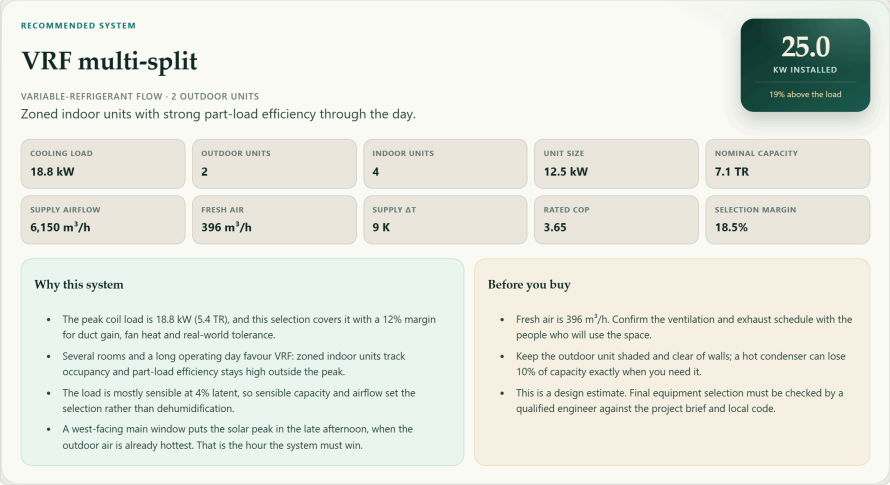


Figure 7 — The recommended system card. A live capture from the studio. The large figure is the installed nominal capacity; the smaller cells are the inputs to the next decision.

4.3 The client view

Below the card, the design day is drawn as a scene rather than a chart. A building, a sky whose colour follows the hour, a sun or moon, lit windows and an outdoor unit change as the time slider moves from midnight to midnight. The panel stays simple: time, outdoor temperature, indoor temperature and one message — comfort held, or comfort at risk.



Figure 8 — The design day at 15:00. The hottest hour: 46.0 °C outside, 24.0 °C inside, comfort held inside the band. The scene is drawn from the same simulation result.

05

CHAPTER FIVE

The hottest day, hour by hour.

A 24-hour simulation built from the same inputs as the recommendation. It answers the question a design load cannot: what does the system do between midnight and midnight?

5.1 Model and operating schedule

The outdoor temperature follows a 24-hour curve that peaks at 46 °C at 15:00 and falls to about 33 °C before dawn. Solar gain follows the chosen orientation, internal gains follow the occupancy schedule, and the zone is a lumped thermal capacitance. The room therefore has memory: heat stored in the fabric in the afternoon is released after the plant stops. The plant holds 24 °C through the occupied day, releases to a 28 °C setback overnight, and pre-cools from 04:40.

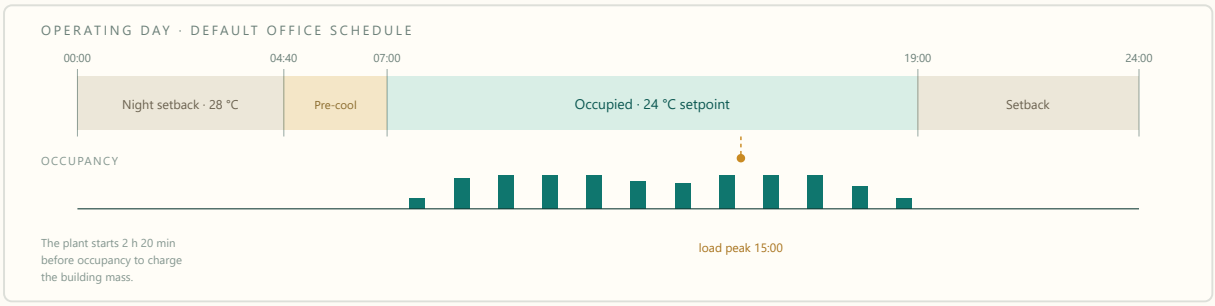


Figure 9 — Operating-day schedule. Four bands: night setback to 04:40, a 2 h 20 min pre-cool, the occupied period from 07:00 to 19:00, and the evening setback. Occupancy bars and the 15:00 peak marker share the setpoint axis.

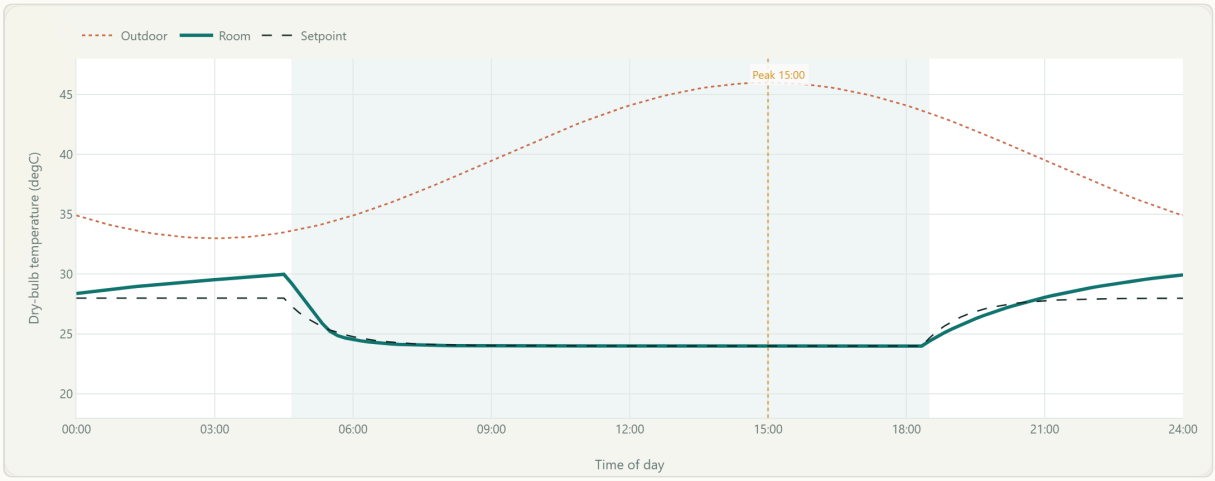


Figure 10 — Zone temperature against outdoor temperature and setpoint. The room drifts to almost 30 °C by 04:30 while the plant is off, is pulled down once pre-cool begins at 04:40, and holds 24.0 °C from about 07:00 until the plant stops at 18:30. The outdoor peak of 46 °C at 15:00 is marked.

Three features deserve comment. The overnight drift is upward although the outdoor air is cooling: the fabric is still releasing the previous day's heat. The pull-down is steep — nearly 6 K in about ninety minutes — because the plant runs at full capacity against an empty room. The occupied plateau is almost flat, the signature of a correctly sized plant under modulating control.

ROOM PEAK	PULL-DOWN	OCCUPIED PLATEAU	OUTDOOR PEAK
30.0 °C	5.9 K	24.0 °C	46.0 °C
04:30, before pre-cool	in about ninety minutes	07:00 to 18:30	at 15:00

5.2 Cooling, capacity and energy

Total cooling is the sum of sensible and latent removal; sensible cooling is the part that changes the zone temperature. The gap between them is small all day because the climate is dry and the internal latent gain is modest, consistent with the 91.6 % sensible ratio of the design load.

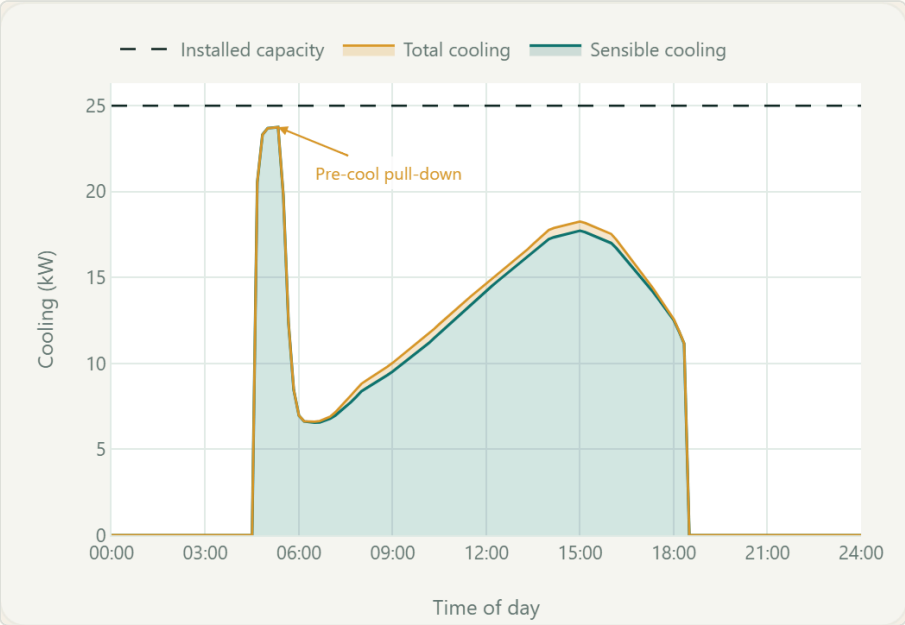


Figure 11 — Cooling delivered against installed capacity. The 25 kW installed line is never crossed. The pre-cool spike at 05:20 reaches 23.75 kW; the occupied afternoon rises smoothly to 18.3 kW at 15:00 and falls to zero when the plant stops.

Indicator	Value	Comment
Peak sensible load	17.73 kW	at 15:00, with the outdoor peak
Peak delivered cooling	23.75 kW	at 05:20, during pre-cool
Peak electric demand	7.47 kW	compressor plus fan
Cooling delivered, 24 h	190.89 kWh	sensible plus latent
Electricity used, 24 h	72.77 kWh	ambient COP correction and fan power
Day-average COP	2.62	below the rated 3.65, as expected at 46 °C
Comfort, occupied hours	100 %	ET* 23–26 °C and RH 30–70 %
Capacity headroom at peak	29.1 %	25.0 kW installed against 17.73 kW
Plant running time	13.83 h	including pre-cool and part-load operation
Room humidity range	**36.7–52.9 %**	**inside the comfort band all day**

Table 7 — Design-day summary. All values produced by the 24-hour run with the default six answers. The day-average COP of 2.62 is a result, not a fault: the rated 3.65 is quoted at 35 °C ambient, and the cycle loses efficiency as the condenser approaches 46 °C.

The 29.1 % headroom is deliberate. It absorbs the pre-cool transient, the ambient derating of the condenser and the difference between nominal and installed capacity. On the afternoon plateau the plant runs at roughly three-quarters of its installed capacity, which is also where a VRF system is most efficient.

> **WHAT THE 23.7 KW MEANS**
>
> The 24-hour run peaks at 23.75 kW at 05:20, above the 19.16 kW design coil duty. The design point is a steady load at 15:00; the simulation also counts the pre-cool, when the room is taken from its 28 °C setback to 24 °C. A plant selected exactly at 19.16 kW would fail on that hour, which is why the recommendation applies a margin.

06

CHAPTER SIX

Built to be checked.

A thin interface over a tested calculation library. Every number on screen comes from a function that can be run, and has been run, without the interface.

6.1 Architecture and stack

The project is split into two layers. The *hvac* package contains pure functions and frozen dataclasses: thermal loads, psychrometrics, the air-handling unit, the refrigeration cycle, the control loop and the 24-hour simulation. It imports no Streamlit and draws nothing. The *views* package and *main.py* contain the interface: widgets, charts, translations and the router. Each module exposes one primary entry point with an input dataclass and a result dataclass, so the hand-off between modules is a real function call.

Stack. Python 3.12.2 · Streamlit 1.50.0 · Plotly 6.3.1 · pandas and NumPy 2.3.3 · CoolProp 6.7.0 · pytest 8.4.2. The thermodynamic properties are the only external physical dependency; everything else is open source and pinned.

6.2 Bilingual interface and theming

Arabic is not a translation layer applied after the fact: the page direction flips to RTL, the sidebar moves to the right, the navigation arrows reverse, and the typography switches to an Arabic face. Numbers, units and refrigerant designations stay left-to-right inside the mirrored layout. A separate light and dark theme changes surface and chart colours through one token set; the two controls give four combinations, all screenshot-tested.



Figure 12 — The Arabic interface. The same studio page in Arabic. The layout is mirrored, the headings are set in an Arabic face, and the tags remain readable as technical labels. Switching language does not recompute anything; the case is preserved across the switch.

6.3 Quality assurance

The calculation library is covered by a pytest suite of 92 tests: algebraic checks against hand calculations, physical invariants (energy balance, monotonicity of load with outdoor temperature, COP below Carnot), boundary and error handling, and regression tests that pin the default case to the values reported here. A scripted browser session captures each page in both languages and both themes and checks that no element overflows.

AUTOMATED TESTS	INTERFACE MATRIX	REPORT PAGES
92	24	14
all passing at the time of writing	2 languages × 2 themes × 6 states	A4, checked page by page

07

CHAPTER SEVEN

The numbers hand off.

Four modules, one chain. Each stage consumes the result of the one before it; the limits are stated as plainly as the results.

7.1 Integrated results

The values below are passed between functions, not copied between pages. Changing an input in Module 1 moves every number downstream. The 18.83 kW and 19.16 kW differ by 1.7 %, entirely because the quick studio assumes 11 occupants and a 46 °C day while Module 1 uses 20 occupants and a 45 °C teaching day.

Stage	Value	What it feeds
Quick studio — six answers	18.83 kW	Recommendation; design-day simulation
Module 1 — detailed load	19.16 kW	Module 2 space load
Module 2 — coil duty	19.17 kW	Module 3 evaporator duty
Module 3 — cycle COP	3.976	Module 4 electricity conversion
Module 4 — comfort run	**24.50 °C · 100 %**	**Design day — 190.9 · 72.8 kWh**

Table 8 — Integrated result chain. If Module 2 is fed 19.16 kW it returns a 19.17 kW coil, a difference of 12 W or 0.06 %. If Module 4 is given the cycle COP it converts 73.2 kWh of cooling into 18.4 kWh of electricity over the eight-hour control run.

7.2 What the model is, and what it is not

It is a single-zone, lumped-capacitance model with a real weather curve, a real load engine and a real control loop. It is not a substitute for EnergyPlus or IES: it has no inter-zone airflow, no three-dimensional shading, no duct thermal mass, no equipment part-load curves beyond the ambient COP correction, and no cost model. Only R134a is available in the cycle module, and the control loop does not model duct transport delay or actuator hysteresis. The next additions, in order, are R32 and R410A, a part-load ratio curve, and a second zone sharing an outdoor unit.

7.3 Conclusion

For the default case the answers are concrete. A 96 m² office in Baghdad with standard construction and 25 % west-facing glazing needs 18.8 kW, or 5.36 tons, of cooling. A VRF multi-split with two 12.5 kW outdoor units and four indoor units covers it with an 18.5 % margin over the selection target. Over a 46 °C design day the installed system holds 24.0 °C and 100 % occupied comfort, delivers 190.9 kWh of cooling and consumes 72.8 kWh of electricity at a day-average COP of 2.62. The detailed chain independently returns 19.16 kW, a 19.17 kW coil, a COP of 3.976 and a final zone state of 24.50 °C.

Those numbers are not the point on their own. The point is that every one of them can be traced to an equation, a preset or a function in the repository — and that a student who can trace them can defend them.

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